Physical Electronics Oven

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Introduction:

In physical electronics class, students need to process silicon wafers to make a variety of electrical devices. To do this, a 500 W halogen bulb (Figure 1) is used to heat the silicon wafers which allows wafers to be doped and contacts to adhere to the wafers (Ekkens 2020). Wafers are doped to change their electrical properties, and contacts are adhered to the wafers so that people can use the changes in electrical properties. This allows students to make devices comparable to those available commercially and at a low cost. The current process requires students to balance their wafers on a cylindrical bulb which is then powered to heat the silicon wafer for an unprecise amount of time. While using the oven the wafers may become unstable due to imprecisions, leading to improperly adhered contacts, improperly doped wafers, and broken wafers.

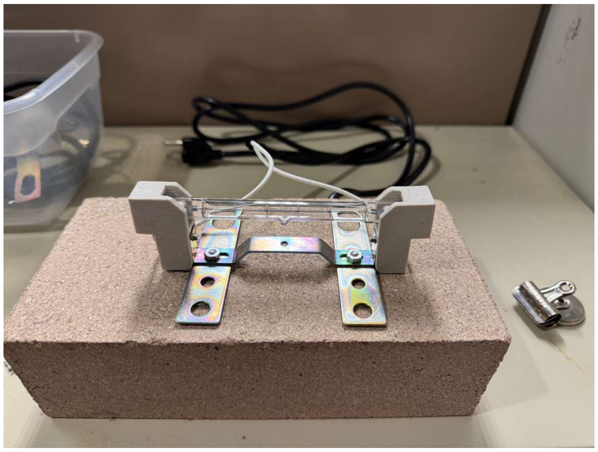

Figure 1: Current oven setup, Maiorov 2025

Dr. Ekkens almost never teaches a lab the exact same way year to year. Each year he will learn something new that will improve the lab or try to incorporate a new element so that students are more engaged. I have seen this firsthand when I take a class from him one year and then become a teacher's assistant for it the following year. When I took physical electronics from him last spring, he added the use of Figure 1 into the creation of our semiconductor devices, and he added the use of a nanocellulite to make better transistors. So, when I came to him looking for a possible capstone project related to physical electronics, he suggested that improvements could be made to the current oven, Figure 1.

Problem Definition

The current rapid heating method, Figure 1, used in the physical electronics class, has unknown temperature, nonuniform heating of wafers, and inconsistent timing. This project aims at improving the wafer processing mechanism by developing an oven for laboratory experiments that is designed for adequate temperature control, stable wafer mounting, and safer day-to-day functioning.

Literature review

Part of the basis of this project is Dr. Ekkens' paper on "An Inexpensive Semiconducting Device Laboratory: Building a Schottky Barrier Diode" which was then adapted into being used in all the physical electronics labs. Dr. Ekkens explains how a student would use the halogen bulb, Figure 1, to adhere metal foil on to the silicon wafers, "The work light should be oriented so it is pointing straight up and the glass face should be rotated out of the way. The wafer should be balanced directly on the bulb with the foil pieces on top. When the wafer is successfully balanced on the bulb, turn on the light for 60 seconds. Fumes from the baking should be vented away from students and into a fume hood if possible" (Ekkens 2020). The technique can also be used to dope silicon wafers. This process is what the project will be addressing by determining the temperature of the halogen bulb to control how long the bulb is turned on and off, and a second bulb is added so that the wafer is more easily balanced.

Onthank et al. is similar to this project due to its focus on affordability. Part of the Open Acidification Tank Controller is being able to control the temperature of a water tank. To do this they are using two relays, one for a heater and the other for a chiller, as "The control aspect of the system is provided by switching on and off standard 110-volt AC power to attached external devices using the 2-channel relay module" (Onthank et al. 2023). For this project, Solid State Relays (SSR) are being used instead of mechanical relays because of their increased efficiency. Another important part of their printed-circuit board (PCB) design is the use of mostly premade modules that attach via solderable through holes. Using parts that are modular and attached through holes helps with manufacturability. Someone building this board would only need a soldering iron, screwdriver, and a drill. This approach to manufacturability will also be when designing this project's PCB.

Prusa is a 3D printer company that is at the forefront of creating open-source materials for the 3D printing community. Because of this, Prusa has developed a robust human interface system using one encoder and two push buttons that allow users to control almost every aspect of their 3D printers. The main display (Figure 2) shows, "Nozzle temperature (actual / target temperature), Heatbed temperature (actual / target temperature), Progress of printing in % - shown only during the printing, Status bar (Prusa i3 MK3S ready. / Heating / file_name.gcode, etc.), Z-axis position, Printing speed (default 100 %), Remaining time estimation" ("LCD Menu (Original Prusa I3) | Prusa Knowledge Base" 2024). A similar display will be used for this project adjusting certain parts, like number 5 and 6, to cater to the needs of the oven.

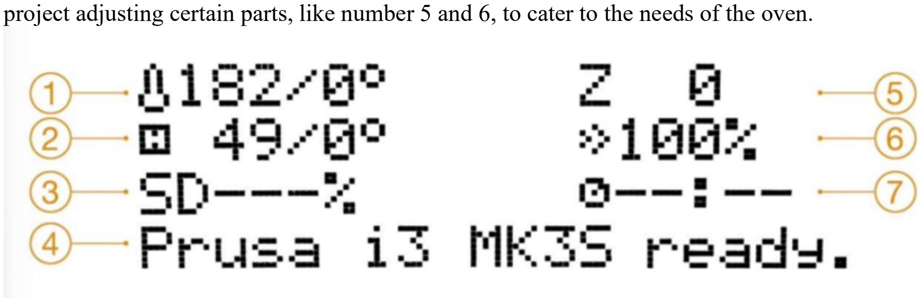

Figure 2, "LCD Menu (Original Prusa I3) | Prusa Knowledge Base" 2024

MicroPython is a programing language that lets a program use Python on microcontrollers. It is a derivative of Python making development easy ("MicroPython - Python for Microcontrollers" n.d.). Additionally, using the Thonny integrated development environment makes it incredibly easy to upload code. It also had dedicated resources for programming Raspberry Pi Pico 2s. Because it is a higher-level programing language it is slower.

Raspberry Pi Pico-series C/C++ Software Development Kit (SDK) is a C++ SDK meant for the Raspberry Pi Pico 2 (*Raspberry Pi Pico-Series C/C++ SDK: Libraries and Tools for C/C++ Development on Raspberry Pi Microcontrollers.*, n.d.). It allows for bare metal programing of the Raspberry Pi Pico 2, and code is faster since C++ is a lower level language. However, getting code to compile requires many complex tools that are prone to bugs.

Standards:

IPC-2221A is a PCB design standard that dictates trace widths and trace separation to ensure that no short circuits happen. It will primarily be used to design the input of power from a 120 V AC receptacle to power converters and to the halogen bulbs. The specific standards are in section *6.3 Electrical Clearance* where table 6-1 establishes the spacing requirements for different board constructions (IPC 2002).

I2C is a popular serial communication standard that allows multiple devices to transmit and receive data on a single data line (*Wikipedia* 2025 "I^2C").

Parallel is another communication standard that will be used to put information on the Liquid Crystal Display (LCD) screen. It sends data over multiple data lines (*Wikipedia* 2025 "Parallel communication").

NIST Thermocouple is the standard set by the National Institute of Standards and Technology for how to read different classes of thermocouples. It will be used to take accurate temperature data from thermocouples to tell the Proportional-Integral-Derivative (PID) control what to do with the halogen bulbs (Burns et al. 1993).

Similar Products:

There are similar commercially produced products that have been considered for this lab. One of which is the generic McMaster oven which meets both the size and temperature requirements of a minimum temperature of 1000 °C and working area greater than 2 cm^2 ("McMaster-Carr" n.d.). But it has no advertised warm-up rate and costs $1700-2000 depending on the size you chose ("McMaster-Carr" n.d.). Another is the RTP-100 Rapid Thermal Annealing Oven, which is a research grade oven that can reach 1200 °C, can accommodate a 10cm wafer, has a warm-up rate of 150 °C/s, and has environmental control (Nano Vacuum Pty Ltd, n.d.). Finally, the MILA-5000 has many appealing properties such as 50 °C/s heating rate, environmental controls (Air, Low Vacuum, Gas flow), temperature up to 1200 °C/s, and a working area of 2 cm^2 *(Mini Lamp Annealer MILA-5000 Series – ADVANCE RIKO,Inc.*, n.d.). Both the RTP-100 and the MILA-5000 do not have pricing on their sites but if something requires you to ask for a quote from them to buy it tends to be expensive.

None of these products fit the criteria of this project, which includes price and heating rate. To expand the availability of semiconductor production within an educational environment, one of the goals of this project is to create an inexpensive and open-source project. Similarly to Dr. Onthank's Open Acidification Tank Controller, this project should be an alternative for low fund labs so they can run semiconductor experiments, without needing to spend over $500 per device. Additionally, to complete a semiconductor experiment within a single lab's timeframe requires that contacts be able to be added in less than a couple of minutes.

Problem Definition:

The goals for this project are to be able to reach a temperature between 1000-2000 °C and be able to sustain it for two minutes at a time. The minimum working area should be 2 cm^2. The heating should be close to that of the current system. For this project, a PID controller will be implemented to control the temperature output of two 500 W halogen bulbs. This will ensure uniform heating of silicon wafers.

Objectives:

The PID controller and embedded system for the project will be made from a Raspberry Pi Pico 2, using either C++ or MicroPython to run code. The user interface will use a 4X20 LCD to give information to the user, and an encoder will be used for input. This oven will be used to add electrical contacts to semiconductor devices and for doping silicon wafers. The bill of materials for this project is shown in Table 1.

Table 1: Bill of Materials, Maiorov 2025

Parts	**Price in USD**	**Sourcing**
RP 2350 Board	7	pipico2
(2x) 500-W Halogen bulb	11	bulb
Bulb holder	10	Ebay or amazon
USB Wall plug	1.67	amazon
Thermocouple	30.20	therm
(2x) Solid State Relays	36.93	DigiKey
LCD (3.3V)	9.76	AliExpress
Power receptacle	4.17	DigiKey
Encoder	0.436	AliExpress
Micro USB cable	4	amazon
PCB + associated parts	10/in^2	PCB
Sum	~152-250	

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Be sure to also add Table 1. Bill of Materials as a caption.

Constraints & Codes:

The constraints for this project are price, parts availability, and ease of construction. For price, the final per unit price should be below $300 dollars. Parts used will be either general parts like resistors or capacitors or prebuilt modules. The PCB will be designed for US based PCB manufacturer, OSH Park. By keeping the size of the PCB smaller the price per board will also be reduced. These constraints limit parts to mostly through hole components; this also helps the final product to be easily assembled. The PCB will also be designed using IPC-2221A to set how wide traces should be and how much distance should be between them.

Impacts:

This project will increase the availability of semiconductor education around the world. By having a cheaper and open-source version of the ovens used in industry to make semiconductor devices, students in smaller engineering and physics departments around the world can also learn how to make semiconductor devices. Having the project be open source also encourages purpose build derivatives to be easily made. For example, someone could use the same electronics but create a real enclosure to test different oxide growths on silicon wafers. Since it is low cost and very modular, if a part breaks, another can be easily acquired to replace the broken one. This helps keep repair low cost and uncomplex.

The Design:

The Physical Electronics Oven is made up of a Raspberry Pi Pico 2, a 20x4 Liquid Crystal Display, a push button rotary encoder, a K-Type thermocouple, a MCP9600 from Adafruit, a AO300A transistor, a PF240D25R Solid State Relay, two terminal blocks, two 500W blubs, a C14 power receptacle, and various basic components. The Pico 2 will be running the oven by taking inputs from the push button rotary encoder which will change the parameters of the oven and start and stop it. The Pico 2 will also display information about the time left for the current oven operation and the current temperature of the oven. The 500W bulbs are controlled by a PF240D25R Solid State Relay which is controlled by a AO300A transistor which is controlled by the Pico 2 with Pulse Width Modulation. The temperature of the blubs is taken by a K-Type thermocouple and then read into the Pico 2 by a MCP9600 thermocouple amplifier which sends the information over the I2C communication protocol. All these components are connected by a Printed Circuit Board (PCB), as seen in Figure 3. The control system for the oven hasn't been finished yet, it needs a characteristic equation for the 500W bulbs. The PCB is almost done but the high voltage power still needs to be routed.

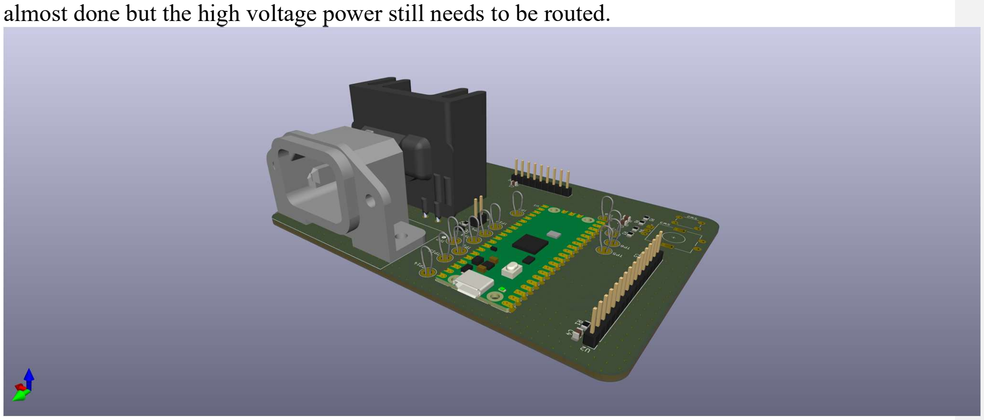

Figure 3: 3D Model of PCB, Maiorov 2026

Using MicroPython a state machine is created where the encoder cycles through available states until the user presses the button on the encoder, which will cause the system to enter that state, Figure 4. After start up, the base state is edit mode. This is where the user can edit important variables such as the temperature, time, and the amount of Proportion, Integral, and

Derivative that is applied to the PID loop. In edit mode, the user can choose one of the variables by moving the encoder to their chosen variable and selecting it with the button and increasing or decreasing the variables by moving the encoder. Once the user finishes editing the value, they press the button which will save the variable and exit back to the edit mode.

Once the user has edited the program, they may start the oven by selecting ‘start’. The oven will do a safety check to make sure the input variables are valid inputs and within the operational range of the oven. If they are not valid, the user will be given a warning and sent back to the edit mode. If the set temperature is too close to 2000 °C or below room temperature, the screen will return an error message. If the time set or any of the PID coefficients are negative then an error stating what value is negative will be returned. If the variables are valid, then the oven will begin running until the timer runs out, or the user presses the emergency stop button. Once either exit condition is met, the oven will return to edit mode. During edit mode the screen will be updated every time an action is done with the encoder or button.

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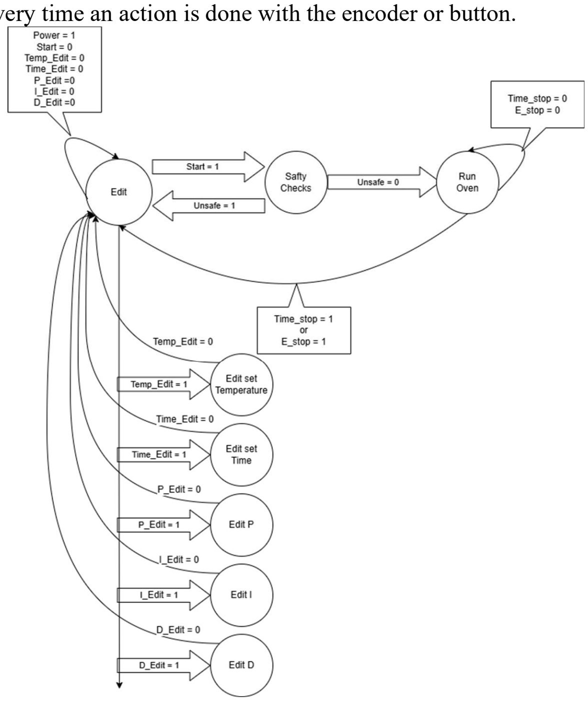

Figure 4: Logic Diagram, Maiorov 2026

The safety check will include ensuring none of the user input values are negative, that the input temperature is lower than the oven can reach, and that none of the PID values are negative. In Figure 5 the code for the safety check shows that it will read the ambient temperature and determine if the set point is too low. While the oven is running the screen will update 4 times per second to keep the user aware of the temperature of the oven and how much time is left for the oven to finish its cycle. As seen in Figure 6, the first line tells the user what the current and set temperatures are and if the user is selecting the ‘start’ mode. The second line gives the time left and set time for the oven, and the ‘Stop!’ option appears only when the oven is running. Line 3 displays what the PID values are, and line 4 tells the user if the oven is running.

```

    if set_temp < KType.ambient_temperature:
        break
    if set_time < 0:
        break
    if P < 0:
        break
    if I < 0:
        break
    if D < 0:
        break

```

Figure 5: Safety Check Code, Maiorov 2026

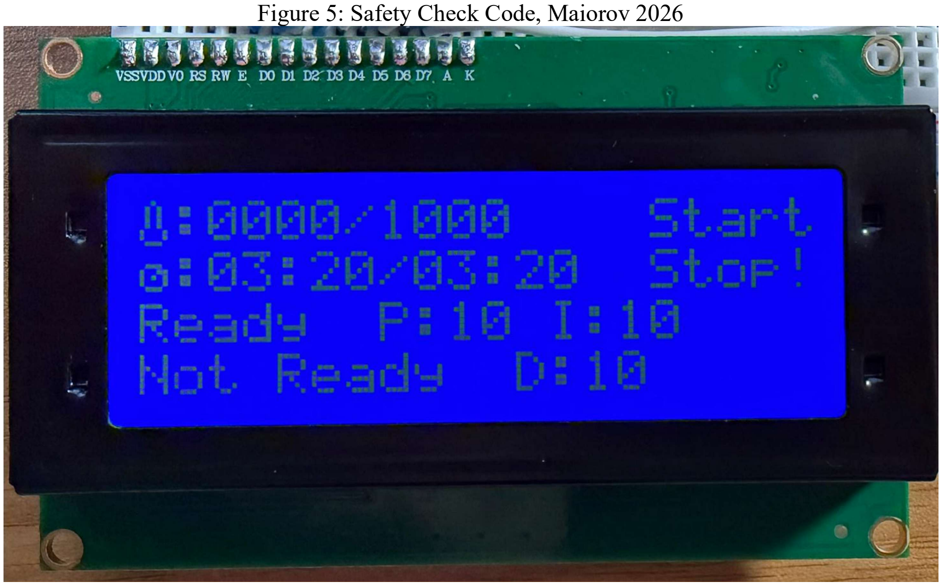

Figure 6: Liquid Crystal Display, Maiorov 2026

This project is designed to allow user interface without needing to alter the software. Alternatively, the user can alter the program directly using software, such as Thonny or another MicroPython IDE, Figure 7. This would require the user to update the input variables within the Python file. In this mode the encoder is only used as an emergency stop. Once the operator starts the oven it will update the screen with the current set temperature and time and update four times a second.

```
main.py

from machine import Pin, I2C
from mcp9600 import MCP9600
import LCD
import mcp9600
import SSR
import time

#Temperature setup
i2c = I2C(0, scl=Pin(17), sda=Pin(16), freq=100000)
KType = MCP9600(i2c)

#Variabes
set_temp = 1000 #degrees C
curr_temp = KType.temperature
set_time = 1000 #IN second
curr_time = 0
P = 10 # Temp values
I = 10 # Temp values
D = 10 # Temp values


LCD.setup(set_temp,curr_temp, set_time,curr_time,P,I,D)

E_Stop = Pin(19,Pin.IN,Pin.PULL_UP)
#Test_LED = Pin(25, Pin.OUT)

E_Stop_down = False

#PWM set up:
```

Figure 7: Simplified Human Machine Interface, Maiorov 2026

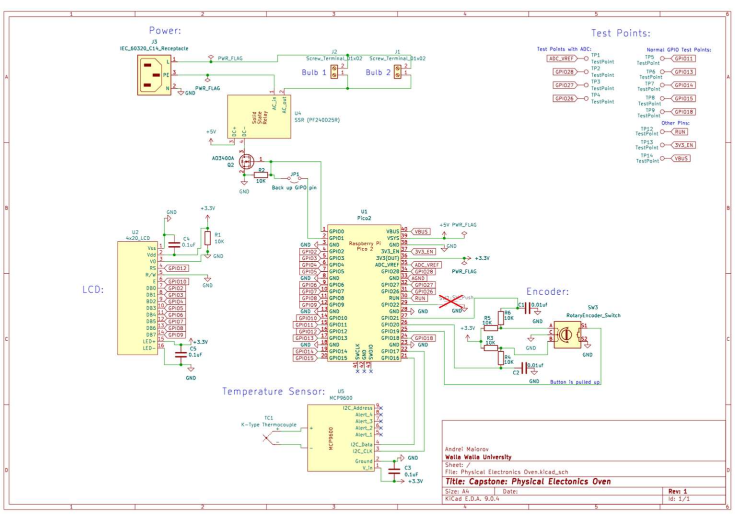

Figure 8: Schematic, Maiorov 2026

Pi Pico 2:

The two parts of this project are the human machine interface and the oven, which are tied together by a Raspberry Pi Pico 2 (Pico 2). The Pico 2 was chosen because of its popularity in hobby spaces and low cost. It also has forty pins in total: eight are ground pins, one supplies 5V power, one accepts 3–5V input power, one is used for an external reset, and one provides 3.3V power. The remaining 26 pins are all general purpose input and output pins with the ability to do Pulsed Width Modulation (PWM), Inter Integrated Circuit (I2C) communication, Serial Peripheral Interface (SPI), and Universal Asynchronous Receiver-Transmitter (UART) communication (*Raspberry Pi Pico 2 Data Sheet: An RP2350-based microcontroller board*, n.d.). The Pico 2 is programed using MicroPython to allow for easy programing. The specific libraries used for interacting with hardware will be discussed in their respective sections.

Human Machine Interface: Display, Encoder:

The human machine interface is made from the Liquid Crystal Display (LCD) and Encoder. This schema for controlling the oven was chosen because it emulates the way the Prusa i3 is set up and used. This version uses a similar 20 x 4 3.3V LCD display and a push button encoder, only missing the “reset button" from the Prusa i3 (“LCD Menu (Original Prusa I3) | Prusa Knowledge Base” 2024). Many companies make 20 x 4 LCDs so many similar data sheets and general purpose programing libraries exist. The one most useful for this project was the POWERTIP PC 2004-A (*Http://Www.Powertipusa.Com/Char.Htm* n.d.). It provides 65 pages of specification for the fully assembled module and each of its components. The pin assignment for the LCD from POWERTIP is seen below in Table 2.

Table 2: LCD Pin assignments, (*Http://Www.Powertipusa.Com/Char.Htm* n.d.)

Pin no.	Symbol	Function
1	Vss	Power supply (GND)
2	Vdd	Power supply (+)
3	Vo	Contrast Adjust
4	RS	Register select signal
5	R/W	Data read/ write
6	E	Enable signal
7	DB0	Data bus line
8	DB1	Data bus line
9	DB2	Data bus line
10	DB3	Data bus line
11	DB4	Data bus line
12	DB5	Data bus line
13	DB6	Data bus line
14	DB7	Data bus line
15	A	Power supply for LED B/L (+)
16	K	Power supply for LED B/L (-)

The display wiring diagrams from the SUNFOUNDERLCD2004 Module wiki were used for prototyping and initial testing of the display. Rather than using an adjustable resistor for the

Contrast Adjust (located at pin 3), a10K ohm resistor is used to maximize the contrast for easier viewing in bright environments, as seen in Figure 6. To program the LCD a generic LCD display driver was added (dhylands n.d.).

For the encoders, a Bourns replica was selected because of its wide availability and near infinite choice of type of ration, switch, and mounting (Bourns, n.d.). The encoder, shown in Figure 9, was connected to the Pico 2 via a breadboard to test functionality. To use the encoder in the program, a quadrature detector is needed. The Pico 2 does not have a hardware quadrature detector, but a generic MicroPython library exists for software quadrature detection and decoding. (Teachman [2018] 2026)

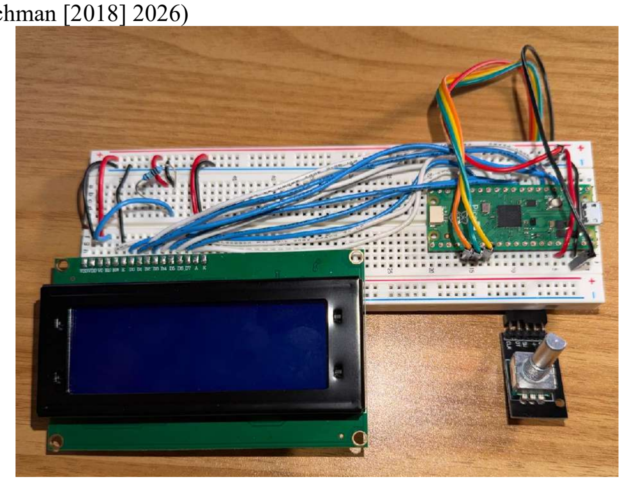

Figure 9: Current HMI prototype, Maiorov 2026

System: SSR, Bulbs, MOSFETs, K-type thermocouple, MCP9600:

The oven consists of two halogen bulbs, a Solid-State Relay (SSR), a Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFET), and a K-Type thermocouple which is read using an Adafruit MCP9600. One Pulsed Width Modulation (PWM) pin toggles the MOSFET which toggles the SSR which then toggles the Halogen bulbs on and off. Depending on how often the bulbs are on versus off will determine the temperature. The temperature is measured using a K-type thermocouple and an MCP9600.

For the SSR, a PF240D25R is being used. It has a rated control voltage of 3-15VDC and current of 15mA. This is on the edge of what the Pico 2 can provide, 3.3V at 12mA, so the MOSFET has been added to use the 5V line to control the SSR. On the load side, the SSR is rated for 12-280VAC at 25A. This should be more than enough for the current heat source and allows for higher current and voltage bulbs in the future.

The SSR is controlled by an AO3400A MOSFET. The AO3400A is a 30V N-Channel MOSFET that is being used to ensure the Pico 2 supplies the correct voltage to turn on and off the SSR. This is achieved by having the Pico 2 drive the gate of the AO3400A with a PWM signal, going from the Pico 2's max of 3.3V down to 0V. Since the AO3400A is an N-channel MOSFET, the source is connected to ground, and the load (the SSR) is connected to the drain. The AO3400A is appropriate for this application because it turns on when a gate source voltage greater than 2.5V is applied.

The halogen bulbs currently being used for the project are 500W T3 R7s Base Dimmable Halogen Light Bulb. The characteristic equation for a Halogen bulb can be found by doing a resistor network heat transfer analysis. The input is the amount of time the bulb is on with 500W of power, and the output would be the surface temperature of the bulb. They are 500W at 120Vrms, so they need the SSRs and PCB to be able to sustain about 5 amps of current. As seen in Ekkens 2020, one of these bulbs plugged in for about a minute is enough to process the silicon wafer. However, the temperature reached by the halogen bulb is currently unknown and uncontrolled.

A K-Type thermocouple was selected because of its wide temperature range, -270 to 1372 °C, and easy availability (Burns et al. 1993). K-Type thermocouples are made up of two alloy mixes, chromel, nickel and chromium, and alumel, nickel and aluminum. This material choice makes the thermocouple very resistant to oxidizing atmospheres, such as air in the atmosphere that the oven will be used in. The thermocouple also has a stainless-steel sheath to protect the wires from the high temperatures. Originally to measure the thermocouple, the Pico 2's analog to digital converter was going to be used. However, this proved impossible, because the Pico 2's analog to digital converter has 12bits of resolution over 0-3.3V, this equals about 0.8mV of precision. This is not enough resolution to read a temperature difference from a K-Type thermocouple because they need a resolution of 41uV. So, to measure the K-Type thermocouple, the MCP9600 by Adafruit was selected because of its ability to work with any thermocouple. It does not have a MicroPython library, but it does have a CircuitPython library. CircuitPython is a branch of MicroPython maintained by Adafruit. But since they are branches of each other a simple MicroPython library can be created to use the basic functions of the MCP9600. So far for this project the thermocouple reading, ambient reading, and differential reading have been implemented. An additional functionality that can be added is the programmability of the alert pins on the MCP9600. The alert pins go high when the thermocouple reaches a user defined temperature. This will be used to keep the oven from going above a specific user-defined temperature.

Printed Circuit Board and Controls:

The preliminary PCB, shown in Figure 11, has been laid out based on the schematic previously shown in Figure 8. The manufacturing constraints are set by the PCB manufacture, which is OSHPark in this case. For this project the four layer board option was chosen, it has 1 oz copper, minimum trace spacing of 0.127mm, minimum trace width of 0.127mm, minimum annular ring size of 0.1016mm, and an edge keep out distance of 0.381mm. These parameters also help in determining trace width from the equations in IPC-2221:

$$\text{Area equation: } A = \left(\frac{I}{k \cdot T_{rise}^{b}}\right)^{\frac{1}{c}} \quad (1)$$

$$\text{Width equation: } W = \frac{A}{t \cdot 1.378} \quad (2)$$

Where $\boldsymbol{I}$ is the current running through the trace, **c, b,** and **k** are constants defined in IPC-2221 that change depending on whether they are inside or outside the PCB for this project they are **c**= 0.725, **b**=0.44, and **k**=0.048, $\boldsymbol{T_{rise}}$ is the allowable rise in temperature during the operation of the oven (currently 10°C), **t** is the ambient temperature of the circuit board (25°C). The current, $\boldsymbol{I}$, running through the traces is found by doing power analysis on the circuit in figure 10. The circuit has an AC voltage source that is 120Vrms at 60Hz with two 500W resistors, the halogen bulbs, in parallel with each other. Since the load is purely resistive the power factor of the circuit is one so the following equations can be used to find the current going through the circuit:

$$\text{RMS to max: } X_{rms} = \frac{X_m}{\sqrt{2}} \; (3)$$

$$\text{Power, voltage, current relationship: } P_{avg} = V_{rms} I_{rms} \cos(\theta_v - \theta_i) \; (4)$$

First the RMS current for one bulb is found with equation 4: P_{avg} = 500W, V_{rms} = 120V. This results in a RMS current of 4.167A, which is rounded up to 5A for factor of safety, its max current, equation 3, is 7.1A. Because the resistors are in parallel the total current that the source needs to supply is 10A RMS, which is 14.14A max and rounded up to 15A for safety.

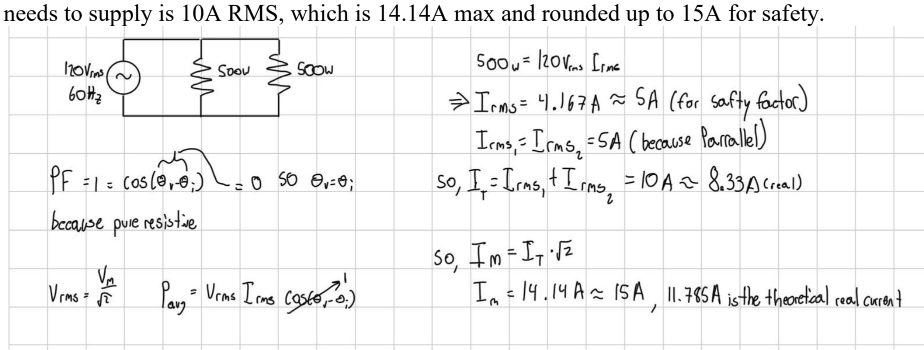

Figure 10: Circuit Analysis, Maiorov 2026

From these current calculations the widths of the primary and secondary traces of the PCB can be calculated, using equations 1 and 2. For the primary traces 15A is the max current, giving a width of 6.3 mm, which is round to 6.5mm. Then for the secondary traces 7.1A is the max current, which gives a width of 2.24mm and is rounded up to 2.5mm. An additional note is that all four layers of the PCB on the high voltage side have the same traces. This so that the traces can use thinner sizes. This is because the current will be spread out across more copper, the suggestion came from Dr. Frohne.

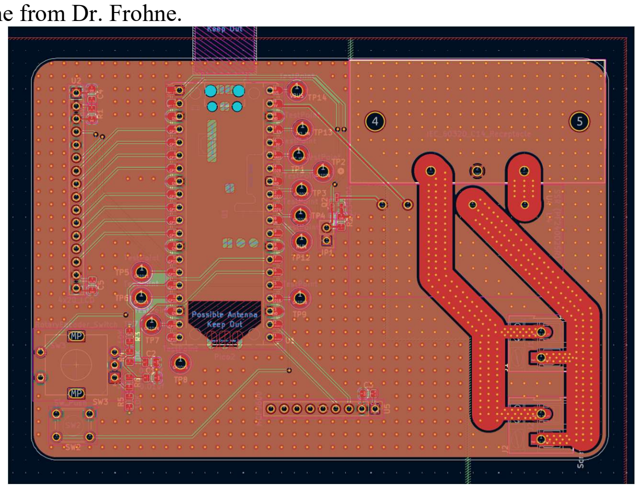

Figure 11: Routed PCB, Maiorov 2026

With the characteristic equation of the bulbs, a PID loop is designed to use the PWM function in the Pico 2 and MicroPython. The final transfer function will be derived from the characteristic equation of the halogen bulbs. This is done by using the following equation:

Transfer function equation: $H(s) = \frac{K(s)G(s)}{1+K(s)G(s)}$ (5)

K(s)is the transfer function of the PID control and G(s) is the transfer function for the Halogen bulbs (*Wikipedia* 2026, “Proportional–integral–derivative controller”). Some early experiments used Juperterlab, a Python IDE, to simulate the loop and begin tuning the PID controller.

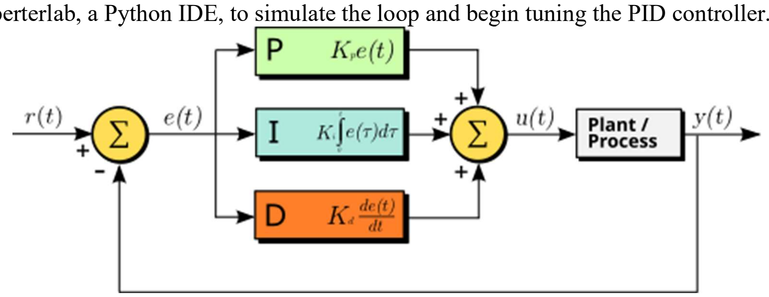

Figure 12: *Wikipedia* 2026, “Proportional–integral–derivative controller”

Bibliography:

Bourns, Inc. n.d. *PEC11R Series 12 Mm Incremental Encoder*.

Burns, G. W., M. G. Scroger, G. F. Strouse, M. C. Croarkin, and W. F. Guthrie. 1993. *Temperature-Electromotive Force Reference Functions and Tables for the Letter-Designated Thermocouple Types Based on the ITS-90*. NIST MONO 175. 0 ed. National Institute of Standards and Technology. https://doi.org/10.6028/NIST.MONO.175.

dhylands. n.d. "Python_lcd/Lcd/Esp32_gpio_lcd.Py at Master · Dhylands/Python_lcd." GitHub. Accessed February 22, 2026. https://github.com/dhylands/python_lcd/blob/master/lcd/esp32_gpio_lcd.py.

DigiKey Electronics. n.d. "PF240D25R." Accessed March 1, 2026. https://www.digikey.com/en/products/detail/sensata-crydom/PF240D25R/753296.

Ekkens, Tom. 2020. "An Inexpensive Semiconducting Device Laboratory: Building a Schottky Barrier Diode." *The Physics Teacher* 58 (1): 64–65. https://doi.org/10.1119/1.5141978.

Http://Www.Powertipusa.Com/Char.Htm. n.d.

info, HEAT-TECH. 2016. "4-2.Types and Mechanisms of Halogen Lamp Gas | HEAT-TECH." July 24. https://heater.heat-tech.biz/halogen-point-heater/basic-knowledge-of-halogen-heater/knowledge-of-halogen-lamps/22776.html.

"LCD Menu (Original Prusa I3) | Prusa Knowledge Base." 2024. November 15. https://help.prusa3d.com/article/lcd-menu-original-prusa-i3_142322.

"LCD2004 Module - Wiki." n.d. Accessed February 22, 2026. http://wiki.sunfounder.cc/index.php?title=LCD2004_Module.

Maiorov, Andrei. 2025. Physical Electronics Oven

Maiorov, Andrei. 2026. 3D Model of PCB

Maiorov, Andrei. 2026. Logic Diagram

Maiorov, Andrei. 2026. Safety Check Code

Maiorov, Andrei. 2026. Liquid Crystal Display

Maiorov, Andrei. 2026. Simplified Human Machine Interface

Maiorov, Andrei. 2026. Schematic

Maiorov, Andrei. 2026. Current HMI prototype

Maiorov, Andrei. 2026. Circuit Analysis

Maiorov, Andrei. 2026. Routed PCB

“McMaster-Carr.” n.d. Accessed November 13, 2025. https://www.mcmaster.com/.

“MECCANIXITY K Type Thermocouple with Plug, Meter Temperature Sensor Probe -50 to 1100°C(-58 to 2012°F) Stainless Steel, 3.2x300mm 3.3ft Wire: Amazon.Com: Industrial & Scientific.” n.d. Accessed March 1, 2026. https://www.amazon.com/MECCANIXITY-Thermocouple-Temperature-Stainless-3-2x300mm/dp/B0D95HB4DV/ref=sr_1_8?crid=1Y3B1QIM0NW0S&dib=eyJ2IjoiMSJ9.hGHryI3k5SF5Ejrs1P-J3qrIqlY9XzWADRuRTiDcmOAuQeDKkk-0-2YqL3CkG7GkZRj48xbjD76Jb3vQAnxI7ST23oprM0iklitNVc_cl-gPAUGLp-3IzOxHIYC6ti79-xYUEQ8dkHR-KzirfadBts307XALSk7EVLRFzYxPLs8F9HectMWACRBLOBoas6lNjlRCw2grYVlGzXpSJNIwO2TmqsBqcXa8jeiIecB5ywg.iHeG36NdnYLps_Ab29HWE6RkGUJShuEQi9QN4dGrc0g&dib_tag=se&keywords=k%2Bthermocouple%2Bprobe&qid=1765472014&sprefix=%2Caps%2C146&sr=8-8&th=1.

“MicroPython - Python for Microcontrollers.” n.d. Accessed November 16, 2025. http://micropython.org/.

Mini Lamp Annealer MILA-5000 Series – ADVANCE RIKO,Inc. n.d. Accessed November 16, 2025. https://advance-riko.com/en/products/mila-5000-series/.

Nano Vacuum Pty Ltd. n.d. “RTP-100 Rapid Thermal Annealing Oven.” Accessed October 29, 2025. https://nanovactech.com/products/rtp-100-rapid-thermal-annealing-oven.

Onthank, Kirt L., James Foster, E. Preston Carman Jr, et al. 2023. “The Open Acidification Tank Controller: An Open-Source Device for the Control of pH and Temperature in Ocean Acidification Experiments.” *HardwareX* 14 (June): e00435. https://doi.org/10.1016/j.ohx.2023.e00435.

“OSH Park Docs ~ Services ~ 4 Layer Prototype Service.” n.d. Accessed March 13, 2026. https://docs.oshpark.com/services/four-layer/.

Raspberry Pi Pico-Series C/C++ SDK: Libraries and Tools for C/C++ Development on Raspberry Pi Microcontrollers. n.d.

Teachman, Mike. (2018) 2026. *Miketeachman/Micropython-Rotary*. Python. October 10, released February 19. https://github.com/miketeachman/micropython-rotary.

Wikipedia. 2025. “I^{2}C.” September 20. https://en.wikipedia.org/w/index.php?title=I%C2%B2C&oldid=1312382986#cite_note-i2c-patent-NL-1.

Wikipedia. 2025. "Parallel communication." September 1. https://en.wikipedia.org/w/index.php?title=Parallel_communication&oldid=1309018209.

Wikipedia. 2026. "Proportional–integral–derivative controller." February 20. https://en.wikipedia.org/w/index.php?title=Proportional%E2%80%93integral%E2%80%93derivative_controller&oldid=1339503842.

Generic Standard on printed board design IPC-2221A. Northbrook, IL: IPC, 2002.

Appendix:

Schematic